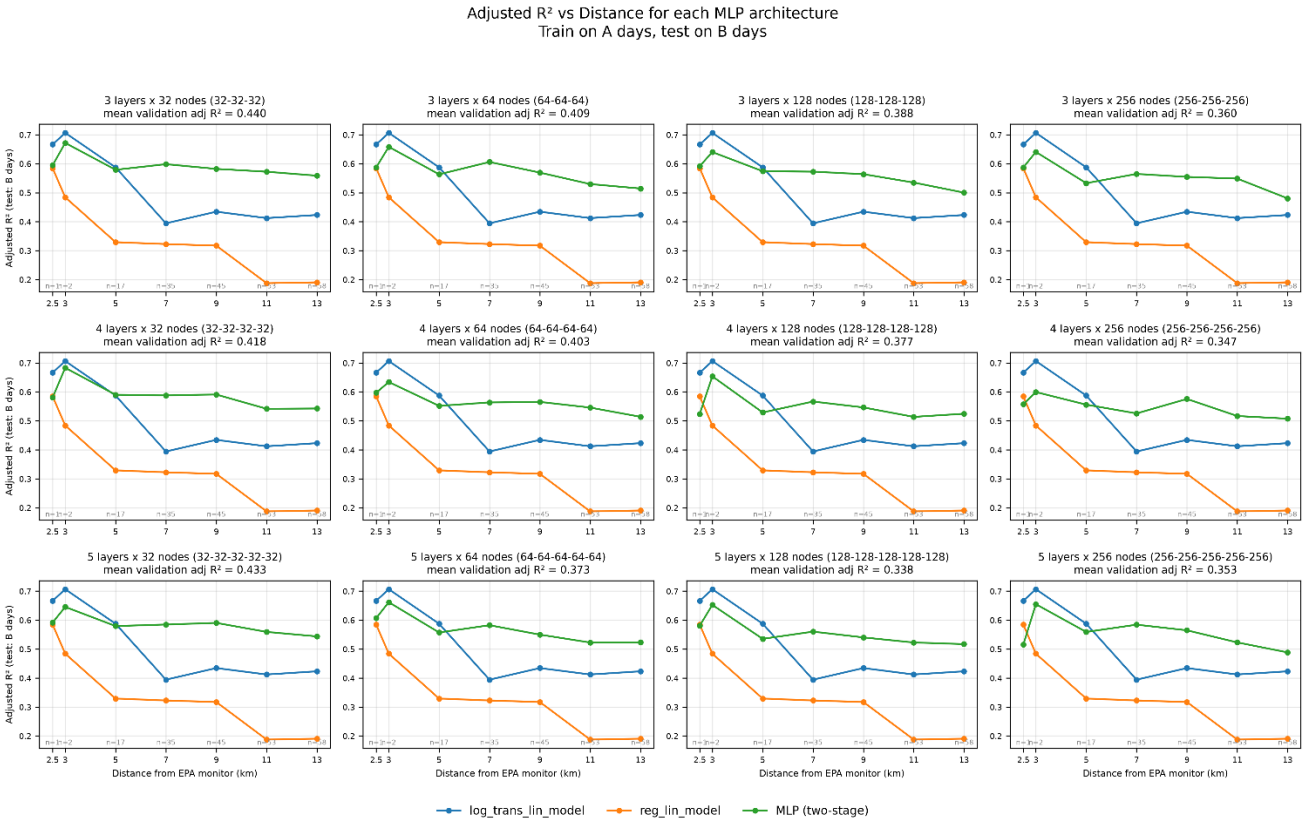
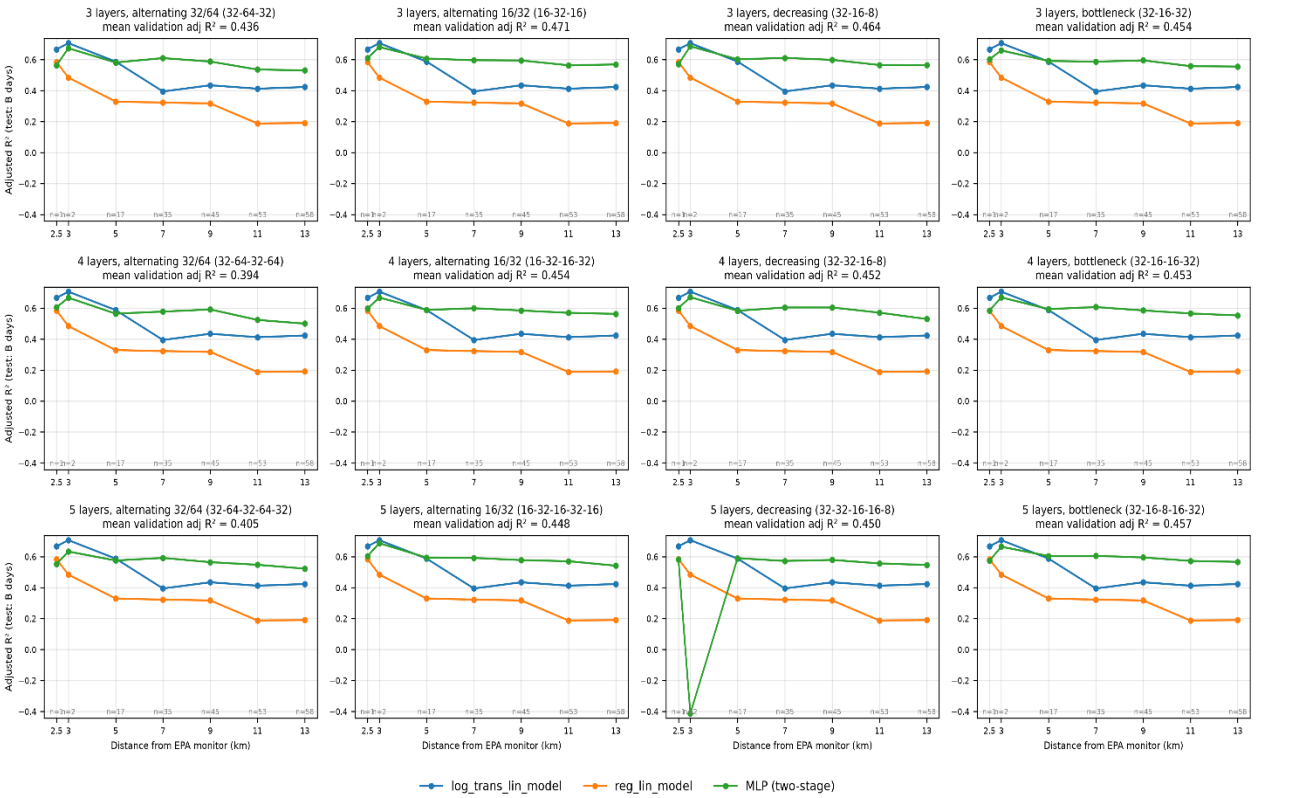
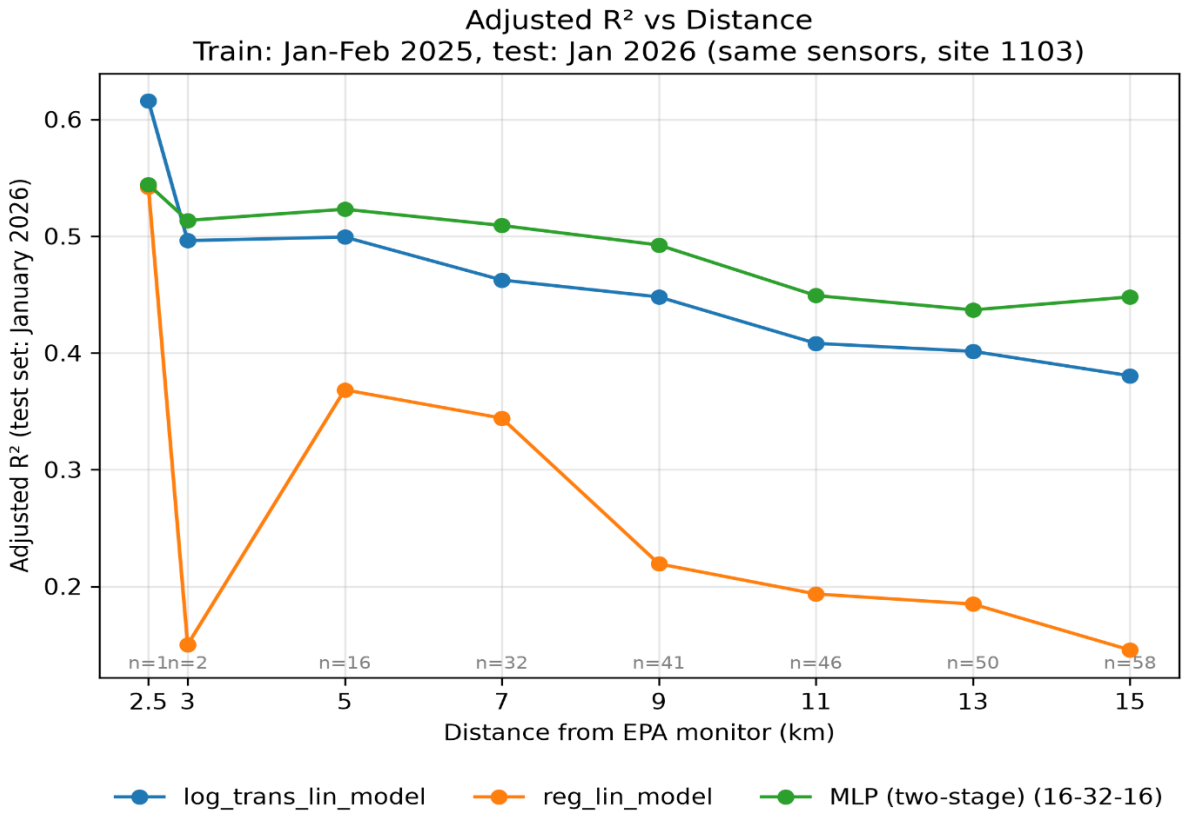
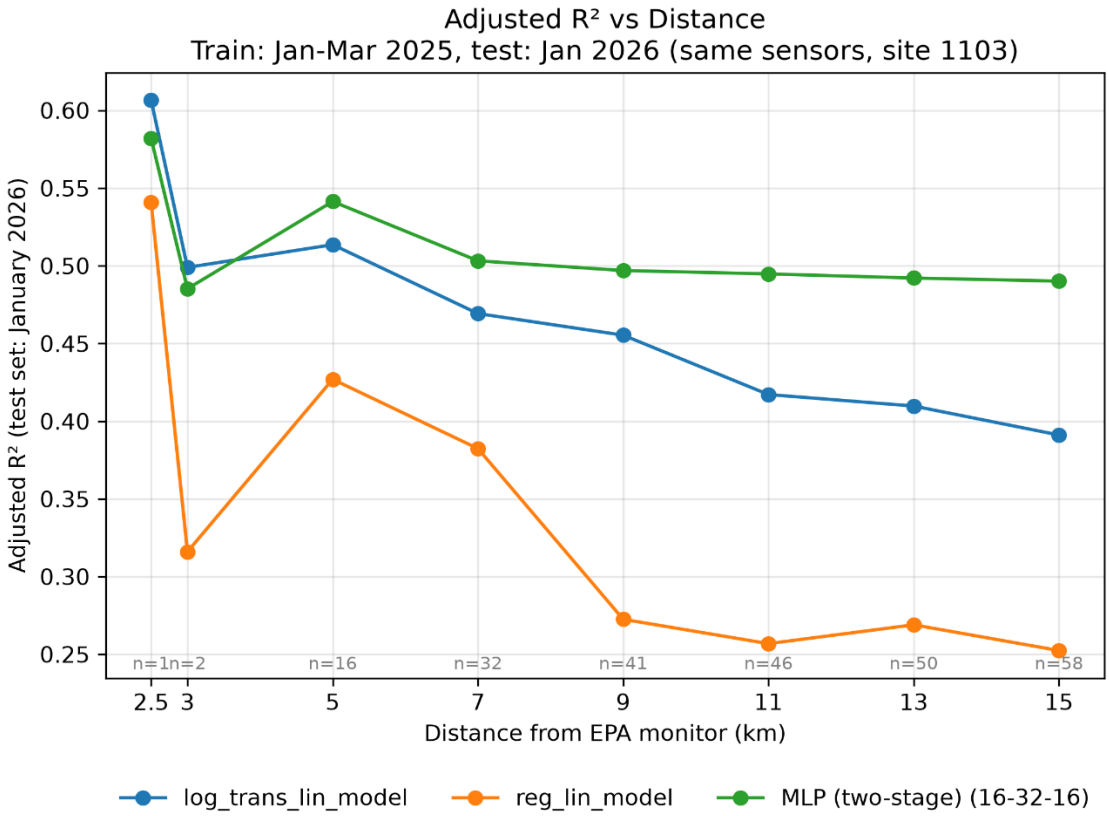
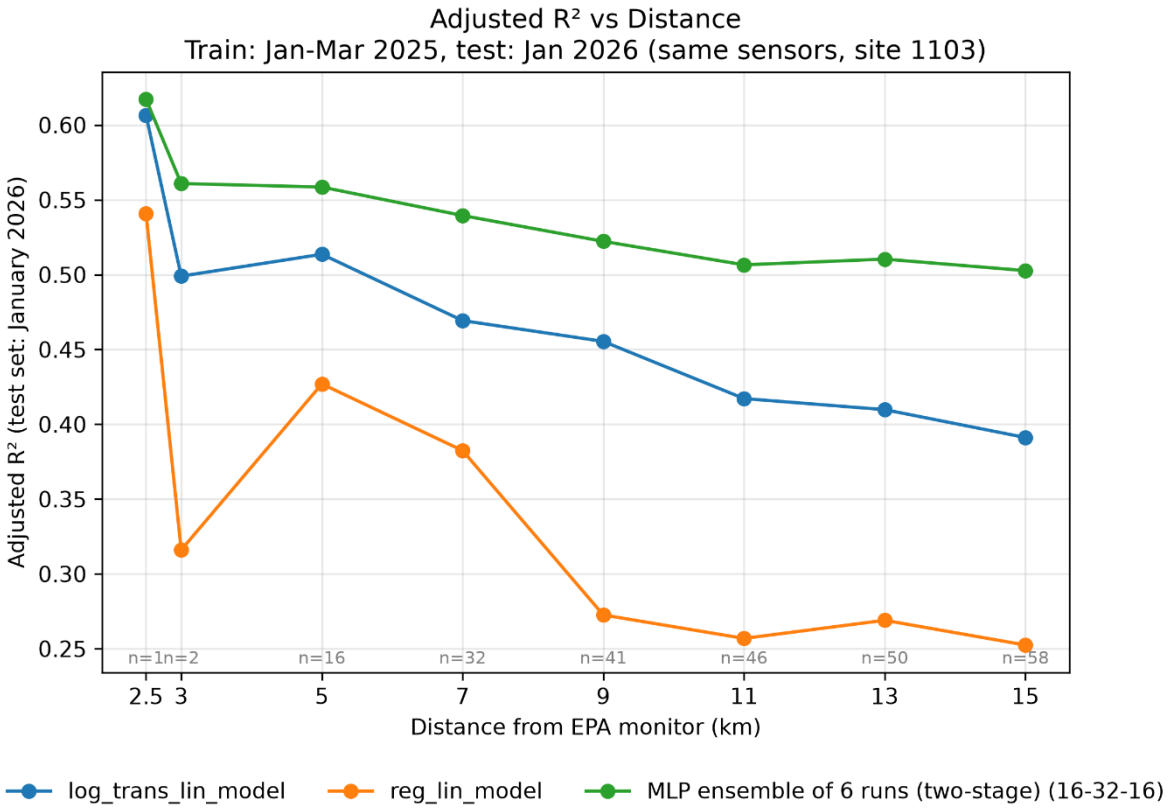

Adjusted R² vs Distance for each MLP architecture
Train on A days, test on B days
3 layers x 32 nodes (32-32-32)
mean validation adj R² = 0.440
3 layers x 64 nodes (64-64-64)
mean validation adj R² = 0.409
3 layers x 128 nodes (128-128-128)
mean validation adj R² = 0.388
3 layers x 256 nodes (256-256-256)
mean validation adj R² = 0.360
4 layers x 32 nodes (32-32-32-32)
mean validation adj R² = 0.418
4 layers x 64 nodes (64-64-64-64)
mean validation adj R² = 0.403
4 layers x 128 nodes (128-128-128-128)
mean validation adj R² = 0.377
4 layers x 256 nodes (256-256-256-256)
mean validation adj R² = 0.347
5 layers x 32 nodes (32-32-32-32-32)
mean validation adj R² = 0.433
5 layers x 64 nodes (64-64-64-64-64)
mean validation adj R² = 0.373
5 layers x 128 nodes (128-128-128-128-128)
mean validation adj R² = 0.338
5 layers x 256 nodes (256-256-256-256-256)
mean validation adj R² = 0.353
Adjusted R² (test: B days)
Distance from EPA monitor (km)
log_trans_lin_model
reg_lin_model
MLP (two-stage)


Adjusted R² vs Distance for each MLP architecture (mixed layer widths)
Train on A days, test on B days
3 layers, alternating 32/64 (32-64-32)
mean validation adj R² = 0.436
3 layers, alternating 16/32 (16-32-16)
mean validation adj R² = 0.471
3 layers, decreasing (32-16-8)
mean validation adj R² = 0.464
3 layers, bottleneck (32-16-32)
mean validation adj R² = 0.454
4 layers, alternating 32/64 (32-64-32-64)
mean validation adj R² = 0.394
4 layers, alternating 16/32 (16-32-16-32)
mean validation adj R² = 0.454
4 layers, decreasing (32-32-16-8)
mean validation adj R² = 0.452
4 layers, bottleneck (32-16-16-32)
mean validation adj R² = 0.453
5 layers, alternating 32/64 (32-64-32-64-32)
mean validation adj R² = 0.405
5 layers, alternating 16/32 (16-32-16-32-16)
mean validation adj R² = 0.448
5 layers, decreasing (32-32-16-16-8)
mean validation adj R² = 0.450
5 layers, bottleneck (32-16-8-16-32)
mean validation adj R² = 0.457
Adjusted R² (test: B days)
Distance from EPA monitor (km)
log_trans_lin_model
reg_lin_model
MLP (two-stage)


Adjusted R² vs Distance
Train: Jan-Feb 2025, test: Jan 2026 (same sensors, site 1103)
Adjusted R² (test set: January 2026)
0.6
0.5
0.4
0.3
0.2
n=1n=2
n=16
n=32
n=41
n=46
n=50
n=58
2.5
3
5
7
9
11
13
15
Distance from EPA monitor (km)
log_trans_lin_model
reg_lin_model
MLP (two-stage) (16-32-16)


Adjusted R² vs Distance
Train: Jan-Mar 2025, test: Jan 2026 (same sensors, site 1103)
Adjusted R² (test set: January 2026)
0.60
0.55
0.50
0.45
0.40
0.35
0.30
0.25
n=1n=2
n=16
n=32
n=41
n=46
n=50
n=58
2.5
3
5
7
9
11
13
15
Distance from EPA monitor (km)
log_trans_lin_model
reg_lin_model
MLP (two-stage) (16-32-16)


Adjusted R² vs Distance
Train: Jan-Mar 2025, test: Jan 2026 (same sensors, site 1103)
Adjusted R² (test set: January 2026)
0.60
0.55
0.50
0.45
0.40
0.35
0.30
0.25
n=1n=2
n=16
n=32
n=41
n=46
n=50
n=58
2.5
3
5
7
9
11
13
15
Distance from EPA monitor (km)
log_trans_lin_model
reg_lin_model
MLP ensemble of 6 runs (two-stage) (16-32-16)